

Data Analysis with NumPy and Pandas

Move from individual Python values to arrays and tables that describe a dataset.



WHY THIS MATTERS

Start from what you already know

Python collections, conditions, loops, and functions

- 01 Explain arrays and shapes in plain language.
- 02 Use DataFrames in a small worked example.
- 03 Complete the guided lab and verify missing values.

CORE CONCEPT

The vocabulary of this topic

Move from individual Python values to arrays and tables that describe a dataset.

arrays and shapes: An array stores same-type values along axes; its shape gives the length of each axis.

DataFrames: A DataFrame is a labelled two-dimensional table whose rows are observations and columns are variables.

filtering and grouping: Filtering selects rows by a condition; grouping partitions rows before applying an aggregation.

missing values: Missing values encode absent observations whose cause and handling can materially change an analysis.

FOUR CONNECTED IDEAS

How the concept works

01

**arrays and
shapes**

02

DataFrames

03

**filtering and
grouping**

04

missing values

NumPy applies one operation across shaped numerical arrays; Pandas labels rows and columns for tabular analysis.

CORE CONCEPT

Read every part of the mechanism

NumPy applies one operation across shaped numerical arrays; Pandas labels rows and columns for tabular analysis.

shape: The length of every array axis.

row: One observation.

column: One measured variable.

WORKED EXAMPLE

Worked example by hand

Calculate the mean of three scores, then group a four-row table by course.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
import pandas as pd
df = pd.DataFrame({"course": ["AI", "AI", "Web"], "score": [70, 90, 60]})
print(df.groupby("course")["score"].mean())
```

OUTPUT

```
course
AI    80.0
Web   60.0
```

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model



One row must have a declared grain, such as one student, transaction, or day.



Axis 0 moves down rows; axis 1 moves across columns in a two-dimensional table.



A boolean mask contains one True/False decision per row and supports transparent filtering.



Missing values require a reasoned policy: retain, impute, exclude, or collect again.

FOUR CONNECTED IDEAS

Notebook readiness map

01

**arrays and
shapes**

02

DataFrames

03

**filtering and
grouping**

04

missing values

Use the concepts from this lesson to read and complete
<https://github.com/curiously/Al-Bootcamp/blob/master/01.python-essentials-for-ai.ipynb>.

RISK REVIEW

Common beginner mistakes

01 Applying an operation along the wrong array or DataFrame axis and misreading the result.

02 Dropping missing rows without explaining what information is lost.

GUIDED LAB

Load a small CSV, inspect its shape, filter valid rows, and calculate grouped summaries.

01

02

03

04

Predict

Run

Inspect

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

- 01 Explain arrays and shapes in plain language.
- 02 Use DataFrames in a small worked example.
- 03 Complete the guided lab and verify missing values.

NEXT EVIDENCE

A notebook that explains row grain, column meanings, filters, missing values, and two grouped results.